The Eerie Machine's Metabolism: A More-than-Human Critical Design Inquiry into the Sonification of Computational Activity

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Within the HCI research field, there is a growing urgency to critically rethink dominant human-centred approaches through posthumanist perspectives in order to reframe responsibility within human-technology relations. Such approaches have been argued to conceal the agential complexity of technological systems; however, posthumanist perspectives often remain distant from broader societal discourses. This study investigates how a speculative sonification of computational processes can shape people's perception and recognition of more-than-human agency in technological artefacts. The research adopts a Research through Design methodology, developing and publicly exhibiting *Contingent Flows*, a Critical Design installation, evaluated through observation and interviews with 19 participants. The findings indicate that perceptions of computational more-than-human agency are mediated by factors such as physical and contextual setting, participants' professional background, pre-existing assumptions about computational objects and specific design choices. The discussion engages with current critical perspectives on seamless design and its implications, the role of human specificity in recognising more-than-human agency and the limitations of the study, while suggesting directions for future research. The thesis contributes to the HCI research field by translating posthumanist perspectives into design practices, offering a speculative mode of engagement with computational systems' more-than-human agency, and evaluating this design approach through the production of empirical and situated knowledge.

Additional Keywords and Phrases: More-than-Human Design, Critical Design, Sonic Interaction Design, Research through Design, Posthumanist Design

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1 INTRODUCTION

Contemporary life, for a large part of the world, is characterised by the ubiquitous presence of everyday computational objects: devices that inhabit our pockets or sit on the corner of a living room shelf, ready to satisfy our needs [14, 33]. Beneath their familiar appearance, however, these seemingly human-tamed objects conceal a more-than-human world constituted by algorithmic activity, data processing and delocalised interconnections [14, 33]. As technology progressively shapes identities, desires, politics and ecological systems [6, 34, 50], the boundaries between humans and technology become increasingly blurred [29, 31].

From a posthumanist perspective, human-centred design approaches in HCI risk backgrounding computational objects' agential complexity through a reassuring, passive facade, supporting their seamless integration into everyday life [14, 33]. This framing normalises the opacity of system operations that accompany users' interactions, within which underlying computational processes remain largely inaccessible while continuing to operate efficiently

across data-driven technological infrastructures [14, 33, 73]. This operational opacity obscures computational objects' participation in broader sociotechnical phenomena such as data capitalism, whose business models depend on users remaining unaware of data processing and extractive mechanisms [14, 33, 73].

In this context, the tendency of human-centred design to reinforce perceptions of computational objects as passive is not neutral but ethically consequential, as it reinforces a tension between how such objects appear and how they computationally operate [14, 33]. This gap conditions how people relate to and act towards technologies, limiting users' ability to question and critically engage with them [14, 33]. Since such relations are also shaped by whether devices are recognised as static objects rather than agential actors capable of producing effects [29, 33], understanding how perceptions of computational agency are formed, negotiated or rejected is ethically relevant for both HCI and society, as it concerns how design mediates the conditions under which more informed and deliberate engagement with technology may emerge. However, while posthumanist perspectives have gained traction in HCI research literature [20, 23], they remain largely unfamiliar in broader societal discourses and face critical limitations regarding their concrete enactment [18, 52].

Building on the Entanglement Paradigm [29], this research adopts a Research through Design (RtD) methodology, developing *Contingent Flows*, a sound-based Critical Design installation exhibited at KTH's Reactor Hall, and studied through observation and in-situ interviews with 19 participants. Drawing on Mark Fisher's Gothic Materialism [25] and the Eerie affect [24], the research proposes sound as an affective medium [35:81–84] capable of rendering computational objects' activity perceivable [55], not by making data processing cognitively understandable, but by making computational presence felt through pre-cognitive affect.

The study addresses a critical gap at the intersection of posthumanist, human-centred and sound practices within HCI. By rendering typically unintelligible computational processes audible, the research aims to speculatively explore whether aurally foregrounding computational objects' activity can prompt an oscillation from human-centred to posthumanist perceptions of such objects, potentially enabling their recognition as agential co-participants rather than passive tools [33], and supporting more aware and intentional engagement with them.

To address the study's aim, the thesis is guided by the research question: "*How can a posthumanist sonification of a computational object's internal processes affect people's perception of it?"* (RQ), addressed through the research sub-questions: *"How can the sonification of a computational object's internal processes be designed to make its agency perceivable?"* (subRQ1) and "*How does the encounter with a sonified computational object reveal and unsettle people's assumptions about it?"* (subRQ2).

The thesis contributes to the HCI field by producing situated insights into how people can perceive and negotiate the recognition of more-than-human agency, while offering a methodological contribution to how posthumanist approaches can inform design practices and generate empirical knowledge.

2 BACKGROUND AND THEORETICAL FRAMEWORK

2.1 Posthumanist Perspectives on Technology in HCI

Posthumanist thinking entails a critical stance towards humanism [26, 39], suggesting that current Anthropocene challenges should be addressed by decentring the human and foregrounding its entanglement with more-than-human entities [64]. In the past decade, posthumanist perspectives have gained greater relevance in HCI research [6, 20, 23], driven by a growing urgency to question the non-neutrality of technological developments and their co-participation in the unfolding of our world [29, 31, 33, 41]. Frauenberger [29] coined this emergent shift as the "Entanglement Paradigm", which calls for a reconfiguration of responsibility through a broader distribution of agency across human and more-than-human actors, enabling a clearer tracing of accountability within sociomaterial assemblages [29, 31, 41].

This emerging fourth wave in HCI is motivated by increasingly blurred boundaries between humans and technology, which complicate questions of transparency, reliability and responsibility [29, 31]. The Entanglement Paradigm gathers various theories holding different visions of agency, object relations and hierarchies [29, 52], but whose shared commitment lies in recognising the ontological inseparability between humans and more-than-humans [29, 41]. In this sense, it frames posthumanist approaches in HCI as complementary [29, 33], rather than substitutes for the third phenomenological paradigm [37], allowing research to explore and integrate critical perspectives "to effectively counteract the technological determinism currently dominating so many of our narratives" [33:44].

While designing technology for efficiency, usability and inclusivity, it becomes urgent to also recognise how its complexity and ubiquitous presence can become a means of structural, productive and invisible power [27, 73]. Giaccardi and Redström [33] stress how seamless human-centred design risks concealing the distributed and operative agency of technological systems, framing them as passive tools. In this regard, a more-than-human perspective enables a critical reframing of the assumptions sedimented in everyday interactions with computational objects, enabled by human-centred framings of interaction [33].

2.2 Sound: An Affective Mediator in Human-Technology Relationships

Sound can be understood as a powerful affective medium. Massumi [49] defines affect as a nonconscious intensity experienced in response to an external stimulus that precedes emotional interpretation, which translates affect into socially recognisable forms without fully capturing its raw intensity [49]. Goodman [35:81–84] proposes a vibrational ontology in which sound operates as an affective material force, acting on the body prior to conscious awareness. Affect becomes relevant for understanding how sound modulates human-technology relationships, not as a carrier of semantic information, but as a pre-cognitive force that influences perception even before conscious sense-making.

Ploeger [55] outlines a historically significant shift in the affective role of Operating System (OS) sounds, tracing their evolution from early computers through the 1990s to the present. The first computers (1940s-1960s) featured high-indexicality sounds, generated by the clicking of their computational components, which provided vital aural feedback for operators to monitor system activity [55, 59]. From the 1990s, tech companies introduced harmonious and pleasant sounds into computational devices to produce positive affect and reduce laypeople's scepticism towards what were perceived as obscure black boxes [55]. Since the 2000s, OS sounds have become increasingly muted [55, 59], shifting their role from the previous reassuring function that characterised the 1990s [55]. As reported in [55], Microsoft in 2015 justified its choice to discard the system initialisation sound with the following statement:

> "We removed the startup sound because startup is not an interesting event on a modern device. Picking up and using a device should be about you, not announcing the device's existence." [70]

As interaction with computers became habitual and socially integrated, auditory cues became unnecessary, allowing sounds to be minimised or removed [55, 59]. Ploeger [55] illustrates how the affective role of sound mediated the evolution of our relationship with technology, leading to its current seamless integration into everyday life while backgrounding its operational states.

Sound practices in HCI more broadly recognise sonification's potential to shape users' emotional responses in technology-mediated interactions through human-centred approaches [17, 28, 38:87–94]. For instance, Sonic Interaction Design (SID) deploys sound to support users' sense-making when interacting with technological artefacts [28, 60], relying on human cognitive processes and familiar, human-scale auditory metaphors [12]. Human-Robot Interaction (HRI) utilises sound to make social robots seem likeable, with the aim of integrating them seamlessly into everyday life [2, 57, 71]. Human-centred sonic approaches afford better sense-making and usability in interaction with technology, alongside their non-exclusive design applications [42, 48]. Nevertheless, they do not

solely regulate emotion, but also mediate people's interpretative framings of computational objects [55]. The tradition of ubiquitous and "calm technology" implies the foregrounding of technology solely in compliance with users' needs [69], corroborating a depiction of technology as reassuring and passive [33], and employing sound, and its absence, as an efficient medium supporting it [55]. However, this portrayal can no longer fully reflect the underlying agential complexity of everyday computational objects, obscuring questions of power, privacy and responsibility [14, 33].

While sound in HCI research often employs human-centred approaches, few studies align with a posthumanist exploration of human-technology relationships, particularly perceptions of more-than-human agency. Wakkary et al. [68] created the Morse Things, bowls capable of communicating among themselves through Morse code, employing a thing-centred approach and analysing how they were perceived by participants in their everyday life. Panariello and Bresin [54] sonified a laptop's software processes, shutdown and idle modes, through an aesthetic sonic mapping based on Glitch Music and the Paetzold instrument. Choi [11] developed a data sonification framework that enables an interactive feedback loop between emergent system behaviour and user actions through visual interfaces, yet it does not empirically investigate perceptions of more-than-human agency.

Empirical research into perceptions of technological agency through sonification remains underexplored in HCI. Therefore, this thesis explores a speculative horizontal extension [34] of current user-centred approaches to the sonification of computational systems, broadening their scope rather than aiming to replace them, drawing on the standpoint that human-centred approaches co-constitute interpretative frameworks of technology as passive [33].

2.3 Ontological and Affective Framing of Computational Agency

The thesis is grounded in Gothic Materialism, theorised by Mark Fisher [25]. This ontology is characterised by immanent forces, woven into the fabric of the world's unfolding, operating through abstract machines (originally conceptualised by Deleuze and Guattari [16]) [25:14]. Abstract machines are nonorganic systems that organise matter through flows, connections and interruptions [25:14]. This perspective engages with systems of power, their flows and their subtle influence on society and individuals [25]. Gothic Materialism allows this research to reframe computational systems' processes not as neutral automated operations but as agential layers of cybernetic machines [25:22, 118], sensitive and self-regulating systems that subtly co-participate in shaping sociotechnical assemblages [25:119]. In fact, Gothic Materialism refuses to separate technology and humanity as distinct domains [25:117–118], conceiving them as co-constituted expressions and collateral sedimentations of late capitalism's abstract machine [25:19], aligning with the Entanglement Paradigm's proposition that humans and things are ontologically inseparable [29, 41].

Of particular significance is Gothic Materialism's account of agency [25] which, drawing on cybernetic principles [25:116], conceptualises it as distributed across abstract machines, in which human and more-than-human entities function as nodes within their articulation [25:14]. As Fisher claims, "[t]o account for agency, we do not have to make reference to any organic or vital at all, but to these machines sensitive to 'disturbances of equilibrium'" [25:116]. Agency denotes not human-like consciousness but rather a system's capacity to register and process perturbations, generating feedback responses [25:22, 115–116]. Thus, agency emerges from the dynamic organisation of systems rather than from a centralised subject [25:15]. This scalable understanding of agency informs the research in two ways. First, it supports understanding computational more-than-human agency as distributed across entangled nodes within abstract machines' activity in broader sociotechnical phenomena (macro-scale agency), whilst also recognising inner computational processes as sites of agential enactment (micro-scale agency). Second, it challenges subjectification by showing that agency is not located in a single entity but emergent within entangled relations according to different epistemological scales. This framework guides the research's speculative sonification approach, in which making computational processes audible exposes technological agency contingently and partially.

The affect through which more-than-human agency becomes momentarily perceptible is what Fisher theorises as the Eerie [24]. It is defined as an affect and mode of being [24:9] that emerges when the locus of agency underlying a phenomenon is unknowable. Fisher articulates the Eerie through two modes: the Failure of Presence (something feels absent where it should be present) and the Failure of Absence (something feels present where it should not) [24:61]. This research focuses on the latter, treating the Failure of Absence as a liminal affect that occupies the threshold between human-centred and posthumanist ontologies, where the sensing of a more-than-human agency precedes its rational categorisation without fully resolving into conscious recognition of its source. In fact, it emerges when sense-making of the agential source underlying a phenomenon challenges our habitual ontological assumptions. Crucially, the agential source must remain unknowable: once it becomes fully intelligible, the affect dissolves [24:62]. For instance, this affect might emerge when interacting with an AI chatbot whose responses feel determined by a more-than-human agency exceeding human understanding, disrupting the ontological categorisation of computational tools as passive [33] by provoking the doubt that there is “something more” at work, exposing one to a form of alterity that is usually not considered possible [24:61–62].

Therefore, the Failure of Absence is employed as a productive framework for rendering the invisible agency of computational objects perceptible, providing a conceptual grounding, translated here into a sonic design approach, that may set the conditions for the sensing of more-than-human computational agency and enable the evaluation of when and how such perceptual shifts occur.

2.4 Sound Epistemology: Listening as Phenomenological Entanglement

The research adopts sound as the primary medium to make computational activity humanly perceivable through real-time sonification of its processes. This choice is guided by sound’s affective quality and its epistemological affordances for engaging with complex systems.

Sound is inherently temporally evanescent and flowing [53:31–32], enabling access to an object’s interiority through a sonic resonant translation of its inner infrastructure, without violating its integrity [53:69–70]. While vision involves stability and the dissection of an object [53:32, 70, 67], sound renders it intelligible through its unifying property, making the act of listening holistic and immersive [53:70–71, 67]. Thus, listening situates the subject within a relational field where knowledge emerges through feeling and entanglement rather than visual detachment [67].

In this research, sound enables computational activity to be experienced as unfolding over time, fostering a relational and attuned entanglement with its source rather than a detached, visually mediated awareness. Through this phenomenological co-experiential mode, computational processes are encountered as they evolve rather than solely represented, allowing processual more-than-human agency to become perceptible rather than merely conceptually acknowledged.

Ananny and Crawford [1] extend this epistemological difference to technological systems. They argue that the privileged models of algorithmic transparency, which imply that "seeing" what is inside a complex system means “knowing”, constitute a limited illusion of control [1:975]. Such approaches are epistemologically insufficient since they often rely on static snapshots of a system, such as source code, while algorithms are dynamic and evolving over time [1:982]. Instead, understanding of complex systems’ processes requires engaging with them in action and relationally, allowing sociotechnical entanglement between human and more-than-human actors to emerge in relation to contingent internal and external factors [1:974, 978].

This perspective supports the research aim of making more-than-human agency perceivable through the entanglement with it, where, rather than being represented, agency perception emerges relationally through the interaction between external factors and its sonic unfolding. Even if the research artefact does not immediately invite an intentional interaction, it remains responsive to it, embodying the potential to gradually make its logic perceptible, constituting a speculative enactment of Ananny and Crawford’s [1] critique.

3 METHODOLOGY

This research is developed within the KTH research project *"Non-verbal communication of invisible information: New methods for communicating complex and inaccessible information through sound"* [75], which investigates sound-based approaches for raising laypeople's awareness of data processing systems underlying digital life.

The thesis explores whether posthumanist sonification of computational processes may challenge assumptions about technological objects as passive tools [33]. To accomplish this, the research adopts a Research through Design (RtD) methodology with a Critical Design approach, developing *Contingent Flows*, a sonic installation exhibited for three days in March 2026 at KTH Reactor Hall, Stockholm. Data were collected through qualitative methods, short-term observations and in-situ interviews, with 19 participants.

RtD addresses "Wicked Problems" in HCI, complex challenges with no single solution [40, 72]. It produces knowledge through iterative problem reframing and design making, rather than linearly applying theory to a pre-defined problem, allowing the mutual shaping of theory and practice through the creation of a situated research artefact [72]. The thesis' abductive approach [58] links theoretical notions of seamless design's depiction of technological passivity [33], sound's affective capacity [35:81–84, 55] and computational devices' decreasing sonic presence [55]. RtD enables this research to bridge theoretical framing and situated empirical knowledge [72], with the installation serving as a probe to articulate and evaluate initial conjectures through iterations between theory, practice and insights emerging from public engagement with the research artefact.

Giaccardi's data-enabled RtD [32] informs this methodology through the recognition of an Agential Shift [32] in the design process, challenging notions of human-centred authorship. Therefore, the sonification of computational internal processes is approached not as a pre-defined representation but as an emergent phenomenon arising from the situated interaction between researchers' intentionality and computational operations and constraints, which mediate and reconfigure access to the system's internal dynamics. In this regard, the RtD process becomes a dynamic negotiation where researchers' intentions, skills and sense-making are continuously reconfigured through the engagement and affordances of the computationally enabled materiality. Thus, knowledge production occurs through an assemblage of co-transformative processes where human (including participants) and more-than-human agencies are entangled in the generation of situated insights [32].

Critical Design challenges inscribed worldviews embedded in design, rejecting the idea that there is no alternative to the status quo of how things are or work [3–5, 19]. Rather than offering solutions, it employs designed objects as media towards the world of ideas, generating new knowledge by leveraging unexpectedness in interaction, which challenges and surfaces the norms underlying society [3]. In this research, Critical Design enables the challenging of normalised framings of technology as passive [33] through a design artefact that leverages ambiguity and defamiliarisation, via the iteration between theory and design choices. Therefore, this study's Critical Design provocation [5] centres on a posthumanist overturning of the seamless, compliant and backgrounded agential activity of computational objects.

SubRQ1 is addressed through the RtD design and making process of the research artefact. SubRQ2 is addressed through the Reactor Hall exhibit study, in which participants engaged with the artefact. The Critical Design dialogical approach [3, 4, 43] positions the researcher as a discussion-facilitator, mediating people's experiences and exploring latent frictions, negotiations and assumptions emerging from their encounter with the installation, capturing phenomenological nuances that quantitative methods would flatten due to their closed-ended format.

3.1 Design Process of *Contingent Flows*

3.1.1 How the Eerie Affect informed Sound Design Approaches

The installation's conceptual design emerged through iterative engagement with Fisher's theory [24, 25] and contemporary art music. Fisher identifies acousmatic sounds, characterised by unidentifiable sources, as potentially

evocative of the Eerie affect, due to the ambiguity or elusiveness of the agential entity producing them [24:81]. The research outlines two sonic approaches to the Failure of Absence within the Eerie affect. While the first aligns with Fisher's examples [24], evoking a Failure of Absence aurally through acousmatic aesthetic qualities crafted by a composer, the second involves systems generating sounds inherent to their own inner workings, with minimal human control over sonic output. In this second case, sound is not Eerie solely because of its ambiguous source, but because it is inherent to what resembles the sonic inner metabolism of a dynamic and responsive system, suggesting purposiveness in what should be a static and passive object. An example is Yasunao Tone's album *Solo for Wounded CD* [63], where the sound is not composed music but the sonification of a CD player's error-correction labour as it attempts to read a damaged disc [15]. Here, the artist becomes mediator, setting the conditions to "give voice" to computational processes rather than controlling their sonic output. The sound produced becomes an externalisation of the system's agential dynamism: inherent reactions to the destabilisation of its equilibrium [25:115–118]. This perspective informs how this research recreates a Failure of Absence within the sonic and systemic design of the installation, where eeriness arises through the perception of an agential behaviour where humans expect passive and predictable machinism. By sonifying computational labour, this approach reveals the autonomous operational forces at work within computational objects that persist independently of human understanding or control, exposing a more-than-human agential purposiveness.

These considerations informed the design process by adopting a sonic palette characterised by sustained, textural and noisy synthetic sound. It was inspired both by Tone's sounds' [63] high indexicality [65], causally linked to the physical conditions of the medium, and early computers' analogue sound [55, 59]. However, the project distances itself from the conceptual stance of Glitch music, emphasising computational existence in its processual activity rather than foregrounding it solely through failure [10, 13].

3.1.2 Data Processing Sonification: Distributed Agency and Pheno-fication

The design evolved from a singular artefact into three interconnected ones, inspired by Gothic Materialism's view of agency as distributed across entities sharing an immanent form of purposiveness [25:15].

Data sonification was guided by the notion of *Pheno-data* [47], framing computational processes as the computer's phenomenological experience. For example, CPU load becomes *Pheno-data*: the system's reaction to processing data, revealing its responsiveness to certain inputs, rather than treating it as isolated and abstract data. Therefore, *Pheno-fication* [47] is used to heighten awareness that behind "perfect-looking red tomatoes" [47:254] (or seamless interfaces) there are complex systemic processes sensitive to change and co-evolving alongside it, "representing the entanglement of the lifeworld" [47:253]. The design process therefore approaches inner computational processes as the material expression of machinic livingness, aiming to make technological more-than-human agency sensorially perceptible rather than explaining how and which data are processed. This theoretical insight further informed the design choice to make the installation responsive to environmental change to foreground more-than-human and human entanglement. Critical perspectives on seamless design [14, 33, 55] were translated into a sonic externalisation of responsiveness based on the system's logic, resulting in a non-straightforward user-centred mapping. These perspectives also informed the decision to minimise the covering of technical components as a response to the visual seamlessness of computational objects, often sustained through aesthetically pleasing designs that conceal their complexity while maintaining a familiar and unobtrusive appearance [14, 33].

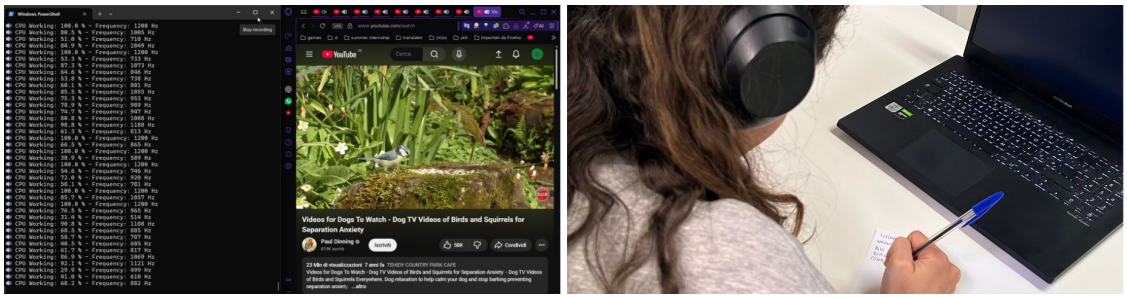
3.1.3 Conceptual Prototyping and Definition of Design Features

The iteration between theory and design practices was supported by the creation of two conceptual prototypes. The first sonified real-time CPU load of a laptop by opening and closing multiple browser tabs with videos playing simultaneously (Fig. 1)**,** constituting the practical application of *Pheno-fication* [47]. The second implemented environmental responsiveness, sonifying the laptop's CPU load through pixel changes detected by the device's

camera (Fig. 2). The latter prototype involved a pilot test with 4 participants, which suggested that people with technical backgrounds may show greater scepticism towards the sensing of more-than-human agency. This insight informed the recording of participants' professional background information and age as contextual data in the subsequent study. Both prototypes used Python code developed with Gemini AI [74].

Through the iterative design process, the main features of the installation were defined:

- Three networked artefacts sonifying inner computational processes linked to proximate environmental changes
- Minimised covering of technical components
- Responsiveness logic privileging computational logic over user-centred mapping
- Continuous sonic unfolding not dependent solely on human presence



Figures 1 and 2: On the left, the first prototype sonifying CPU load; on the right, the second prototype shown to a participant during the pilot study.

3.1.4 The Installation: Contingent Flows

The installation comprises three networked artefacts: one Parent and two Children (Fig. 3). The computational hierarchy and physical appearance emerged through negotiation between human skills, temporal constraints and material availability (see A.1). The artefacts' physical structure (Fig. 4, 5) was inspired by Seznec and Campo Woytuk's *The Period Instrument* [61] and Morozov's *floating point number* artwork [51].

The Parent includes two sensors (camera and microphone), one speaker, one audio interface and one laptop. The two Children are structurally similar: each has two speakers, one amplifier and one Raspberry Pi (Pi3 or Pi5). The Parent differs visually from the Children by featuring a single speaker mounted on a pole, whereas the Children are built on two retort stands with two speakers each, to which components are assembled through clamps and transparent acrylic supports.

The installation system was conceptually designed by the author, informed by theory, and implemented through sonic and technical development by Maurizio Berta (PhD student at KTH). The technical functioning centres on the Parent's capacity to gather environmental data through its light-adaptive camera and microphone, which are respectively sensitive to light and sound variations. The Parent's sonic output depends on the intensity of system calls generated by the analysis of the camera's adaptive response to light changes and by the machine listening process.

The two Children each receive the Parent's analytical data, every time a loudness threshold value is exceeded, via OSC (Open Sound Control) protocol. The Children sonify their own audio production processes, which are steered by the Parent's machine listening analysis, following an analogous sonification logic based on system calls density.

The three artefacts' sounds feed back into the Parent's microphone, creating an interdependent loop between environmental stimuli and the installation's own sound output. Each artefact produces sounds related to its

computational processes, whose sounds are also shaped by their three different operating systems' constraints. Sound variation depends on whether system-call intensity exceeds a certain threshold, with greater stimuli producing more sonic variation. The sound remains continuous, but its unfolding is shaped by environmental stimuli, to which human presence can contribute but is not fundamental.



Figures 3, 4 and 5: On the left, the final setup of the three artefacts exhibited in the Reactor Hall; in the middle, the initial low-fidelity prototype of the artefacts; on the right, a detailed view of one of the Children artefacts.

3.2 Data Collection Methods and Research Study Design

The research study was conducted through the public exhibition of the installation (Fig. 6), using two qualitative ethnographic methods: short-term observation and in-situ interviews [7:34–38, 44–46], with 19 participants recruited opportunistically by the researcher during the event. Participants were aged 21-67 and included 12 students and 7 workers, with backgrounds across STEM, design and humanities-related disciplines. The short-term observations lasted between 2-5 minutes, while the interviews lasted approximately 16 minutes per participant. Participant gender and nationality were not collected in accordance with data minimisation principles in research ethics, as they were not considered relevant to the research questions given the study's qualitative focus on perceptual responses to the installation and its limited sample size.

Since the design of *Contingent Flows* leverages the notion of the Failure of Absence affect [24], which requires an unprimed encounter to subsist, the study was designed to preserve ambiguity and avoid cognitive pre-framing. The public installation context enabled a more spontaneous encounter than a lab setting would have allowed [44], while also allowing in-situ recruitment of participants with greater heterogeneity in age and professional background than would have been possible within the researcher's immediate academic network. Only single or paired visitors were included to preserve natural social dynamics and enable in-depth observation within the practical constraints of a single observer.

Short-term observations were performed during participants' first encounter with the installation. The observation aimed to investigate the affect and the engagement modes enabled by the installation [49, 62], allowing triangulation of the gathered data with the following interviews [44]. The observation leveraged a protocol developed according to the research questions [7:34–38, 44], focusing on spatial and bodily behaviour, facial and emotional cues, and verbal and non-verbal reactions (see A.2).

After the encounter, participants were invited to exit the installation's room to discuss their experience with the researcher. The interview was structured in "before" and "after" the "revelation moment", a specific stage in which it was disclosed that the real-time sound of the installation was related to its inner computational processes and

responsive to environmental stimuli (see A.3). This structure allowed for separating the data in relation to the researcher's mediation over participants' own sense-making about the installation. The first part included explorative questions to elicit participants' unmediated feelings, thoughts, assumptions and observations related to the installation. The second part was guided by a Critical Design dialogical approach, where the researcher assumed the role of facilitator in supporting participants' reflections regarding the research topic [3, 4, 30, 43]. The final part of the interview also allowed participants to transfer their reflections to everyday computational objects, where the sonic behaviour of the installation served as a design probe to support exploration of how they would feel if their everyday devices sounded in a similar manner and whether their perception and relationship with them would change. The abductive nature of the research led to interviews combining semi-structured and unstructured questions, enabling deductive responses informed by inductive insights in relation to the research questions [44, 46]. This qualitative method was fundamental in supporting the explorative approach of the study, affording an understanding of participants' sense-making over time and the different angles through which the installation was interpreted and experienced.



Figure 6: The Reactor Hall setting. On the left and in the middle, two different perspectives of the hall; on the right, the entrance to the installation room.

3.3 Data Processing and Analysis

From the research study, two types of data were collected: written observational notes on paper and recorded interviews. Interviews were transcribed using The European Commission's Speech-to-Text tool [21] and verified through re-listening to the audio recordings.

Data were analysed using an abductive Thematic Analysis (TA) approach [22], following Braun and Clarke's framework [8, 9], which supports the identification of systematically meaningful patterns in relation to the research question(s) within a qualitative dataset [8, 9]. Its flexibility enabled a multi-dimensional analysis, combining and triangulating interview data with observational insights and divergences, and supporting the temporal positioning of interview quotes in relation to the revelation moment through the creation of "before" and "after" codes, contextualising participants' sense-making over time.

The analysis process (Fig. 7) followed an iterative triangulation between observational and interview datasets. It began with familiarisation with the data, providing an initial understanding of meaning tendencies and framing each participant's experience. Raw coding was conducted on a local version of Taguette [56], combining inductive codes grounded on participants' quotes with deductive codes informed by the research's theoretical framework, including the Eerie affect [24], critiques of seamless design approaches [33, 55] and the affective role of sound [35:81–84, 49,

55]. After refining the codes, a series of themes started being hypothesised. To visualise networked clusters of codes [8], the coding was transferred and reorganised to Figma [76]. Once themes were identified, the writing process itself constituted a further analytic stage through ongoing triangulation across data sources.

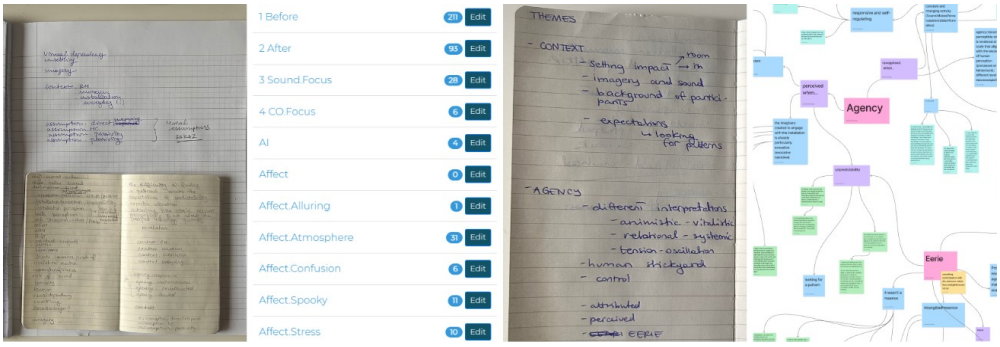


Figure 7: Stages of the data analysis process.

3.4 Ethical Considerations

Prior to entering the installation room and beginning the observation, participants were informed verbally about the contents of the informed consent form, given time to read and sign the paper document (see A.4). Detailed information about the research aim was provided after the interview to avoid influencing participants' responses. Data were pseudonymised and stored on the researcher's password-protected personal devices.

Ethical risks related to participant discomfort, due to the research's Critical Design approach, were mitigated through the iterative design process. Pilot testing informed adjustments to the installation's sound design, specifically avoiding acutely unpleasant sounds to steer the experience towards productive unease rather than sensorial distress. This risk was further addressed by emphasising the voluntary nature of participation and the right to disengage at any time during the in-situ study.

4 RESULTS AND ANALYSIS

This section presents the three main themes derived from the qualitative data analysis. Firstly, it addresses how certain identified factors mediated participants' encounters with the installation, including setting and individual aspects. Secondly, it examines how participants' unsettlement was primarily driven by unpredictable sonic behaviour, and how the urge to identify patterns generated expectations of direct-mapping responsiveness, eliciting different mechanistic, animistic and cybernetic perspectives regarding computational objects' agency. Finally, it illustrates how a hypothetical sonification of inner computational processes in everyday devices may shape perceptions of their underlying activity and overall presence.

Participants' perceptions of the Parent and Children artefacts concerned visual and sonic qualities and were highly subjective and association-based. They were coded but not developed into an analytical theme as they did not reveal consistent, meaningful patterns or hierarchies across the dataset and did not support analytically relevant distinctions between the artefact types. Engagement was exploratory and holistic, with participants shifting between artefacts while primarily referring to the installation as a whole.

4.1 Situated Conditions in Shaping the Emergence of the Eerie

4.1.1 The Reactor Hall as a Contextual Mediator

The unsettlement experienced by most participants was significantly shaped by the Reactor Hall setting, which played a grounding role in mediating the encounter with the installation. The environment's unfamiliarity destabilised habituated experiential frameworks in participants (e.g., P7: *"[…] We're going very down, and compared to the rest of KTH campus, this looks very, very, very distinct and different"*), producing a sense of atmospheric unease interpretable through Fisher's Failure of Presence: a space that evokes eeriness through the traces of a past agency no longer active, leaving visible solely its traces [24:62–63]. P8 stated *"It's like an empty industry and nobody living here. And you just walk through it and listen to some voice up here. [...] Listen, like the sound. The soundscape of the industry"*, while P6 claimed *"Yeah, it feels like a techno rave, but without the techno"*, both echoing the sensing of previous activity's remnants, making visible Fisher's affective structure generative of the Eerie [24].

More than half of participants recognised that the setting's newness, underground location, size and lighting influenced their experience, as P7 summarised: *"I think the aesthetics of the whole being in a nuclear reactor hall definitely add to the experience"*. When a few participants imagined the installation in a neutral setting such as an art museum, they hypothesised it would have felt *"lighter, easier, smoother"* (P16) and *"probably less spooky"* (P15), further confirming the Reactor Hall's role in heightening atmospheric eeriness. This suggests that participants encountered the installation already primed for uneasiness and interpretative uncertainty, where the installation's unsettling design operated within a broader affective context.

The setting also shaped the imagery through which participants framed their experience, evoking Sci-Fi, postapocalyptic, space, horror and music event scenarios. For instance, P5 described feeling *"picked up by an alien ship […] in the dark somewhere and hearing like strange noises"* linking it directly to the enclosed installation's room setting: *"I think it's related to the craft. Like the spacecraft or like the bunker, the building. Being like inside something"*. This shows how the atmosphere of the setting also shaped the listening experience, where descriptions of the sound often reflected the imagery constructed by participants. Sounds were described primarily as machinic and electronic, but also as organic or with similitudes linked to the description of the atmosphere realm (such as aliens, zombies, submarine, spaceship, techno clubs). For example, P5's evoked imagery was also reflected in their interpretation of the sound: *"Yeah. that bop bop bop. That was my alien"*. The same correlation between setting, sound and evoked imagery also emerged in 6 other participants (P6, P8, P9, P13, P15, P17), suggesting that the setting's Eerie atmosphere operated as an interpretive lens shaping not only how the space was felt, but how its sounds were perceived.

Overall, this finding highlights the situatedness of both the research study and the evoked Eerie affect, where engagement was mediated by an assemblage of contextual factors specific to the exhibition setting rather than shaped solely by the installation's design features.

4.1.2 Participants' Backgrounds Influence

Some participants' backgrounds further shaped their interpretation of the installation, influencing how unfamiliarity and unsettlement were experienced.

Most participants focused on the installation's sonic qualities. However, the backgrounds of P11 (electronic engineer) and P20 (mechanical engineer) uniquely differentiated their encounters with the installation from those of the other participants. P11 approached it primarily through a technical and analytical framework rather than an experiential one (*"[...] I was very interested in the physical part of it [...] I wanted to check if this was an amplifier, a computer"; "One needs to kind of try to make sense of what's happening instead of just, you know, be there and absorb it directly"*), focusing on technical implications and their relation to sound. P20's disengagement instead

stemmed from a greater interest in the Reactor Hall as a construction site, focusing on the leftover structural materials rather than the installation itself ("*Oh, the installation. [...] No, I'm just thinking there's a lot of equipment that was in this place. It would be interesting to know a little bit more about that"*). These were the only participants who described their encounter as *"challenging"*, suggesting that their unsettlement derived from background-related expectations about context (P20: *"It's like, oh, what is this here, you're challenging yourself, you know, a little bit"*) and the legibility of system technical functioning (P11: *"It's just, I would say, challenging. One needs to kind of try to make sense of what's happening"*), rather than the Eerie affective unsettlement described by others.

Other participants' backgrounds also shaped the direction of the interviews. For example, P15, an anthropology student, reflected on the social construction of agency (*"[...] the difference is probably just more how complicated it is, how we perceive it [...]. If we agree that something is sentient or has will, then it does, and if we don't think that it has, then it doesn't"*), while P17, a psychologist, connected their unease to sound's psychological impact.

Overall, individual disciplinary lenses mediated some participants' sense-making, highlighting the situatedness of the study and shaping both the emergence of the Eerie affect and interpretations of the encounter with the installation. For instance, for P11 and P20, ambiguity was experienced less as affective unsettlement and more as technical and contextual misalignment.

4.2 Expected Interaction and Negotiations in Recognising More-than-Human Agency

4.2.1 Unintelligible Sonic Behaviour and the Emergence of the Eerie

Curiosity was the dominant emotion across the dataset, often intertwined with unsettling emotions including weirdness, confusion and tension. This unease mainly emerged from the unpredictability of sonic behaviour and participants' difficulty in identifying a stable interpretative pattern (e.g., P13*: "It was draining for me to listen to it [...] because it was messy and [...] I couldn't really see the pattern"*). Despite this, curiosity was sustained by the ambiguity of interaction (e.g., P2: *"[...] suddenly the sound stopped for a while [...] and then I became more curious about whether there was a particular reason behind it or whether I was making movements that caused it […]"*), sometimes supported by visual cues present in the installation. However, participants' attention was primarily directed towards sound, backgrounding computational objects' presence (e.g., P12 *"I think the focus for me was [...] more on the sounds. I really did not pay as much attention to the objects"*). For instance, only a few participants (P1, P5, P6, P11) correctly identified the Raspberry Pis when prompted by the researcher.

Responsiveness was not assumed by all participants: some interpreted the installation as a sonic sculpture rather than an interactive element, without attempting to interact with it. In fact, expectations of passive spectatorship (P5: *"I felt like that was the main thing of the room. So, we were just standing."*, P11, P12, P13, P20) and assumptions about forbidden touch (P16, P17: *"I didn't think that I could touch them"*), common in art exhibition contexts, also shaped modes of engagement.

However, more than half of participants expected the installation to be responsive due to its dynamic sonic behaviour, looking for a direct mapping between sound and their movements. The tension between participants' attempts to find a stable interpretation and the installation's resistance to legibility set the conditions for Fisher's Failure of Absence [24] to emerge. By displacing human-centred expectations of responsiveness, the installation foregrounded a seemingly independent logic (e.g., P10 referring to sound: *"[…] it was doing its own thing, it was not directed at me"*) that participants struggled to interpret. Some participants' intense search for interaction revealed friction in conceiving purposive more-than-human behaviour as unlinked to their presence. As P7 explained *"[...] You feel like you must influence something. But maybe that's not true at all. But it would be nice to think you are influencing it"*. This quote reflects uncertainty about the installation's responsiveness and a desire to maintain a sense of relationality with its sound, suggesting that perceived interaction also functioned as a way of making sense of ambiguous sonic behaviour while sustaining a sense of involvement despite this uncertainty.

For some participants, this ambiguity was affectively experienced as the presence of an unseen agency, producing an ambivalent mixture of attraction and fear, summarised in the code *"It wasn't a menace"* (inspired by P4: *"Alluring. It pulls you in. [...] there was this chaos that was getting closer and closer. I didn't feel like it was necessarily a menace"*). Other participants stated: *"There is something more than just us. [...] But it wasn't frightening"* (P17) and *"I got this eeriness, that was the first feeling, [...] but then I felt attracted"* (P10). These quotes reflect Fisher's notion of Failure of Absence [24]: the installation's encounter aroused both dread and fascination since it exceeded "standard perception, cognition and experience" [24:8], exposing participants to a new form of alterity, where a mysterious agency was sensed, challenging their habitual ontological frameworks, according to which objects are expected to feel passive. This dynamic was also evident after the revelation moment, as some participants' unease decreased. For instance, P17 explained: "*For me it's now less disturbing"* because *"I have an explanation and I know what it is"*, empirically reflecting Fisher's notion that the Eerie dissolves once its agential source becomes intelligible [24:62].

Some participants perceived sound as emerging from an unintelligible purposiveness at work [24:132], through bodily and sensory attunement with the installation's sound. However, the interview questions led some participants (P3, P4, P5, P17) to reframe their encounter through an analytical rather than phenomenological lens, concluding that the sound was recorded. For instance, P3's intense bodily attunement to the rhythm produced a sense of co-dependent relation with the installation, projecting purposeful agency onto it (*"It kind of wanted me to do something"*, *"There's a certain type of beat that follows me"*). However, when later asked whether the sound was real-time or recorded, P3, surprised, hesitated and then concluded it was recorded. This suggests that some participants may initially experience the installation as affectively agentic, before reframing their encounter within familiar and analytical models.

4.2.2 Cognitive Negotiations in the Sense-Making of Interaction

Some participants, despite the perceived ambiguity of interaction or the observed attempt to test the sensors, denied in the interview the installation's responsiveness due to the disrupted expectation of direct sonic feedback which often led them to conclude that the sound was recorded (e.g., P2: *"At first, I thought my movement was reflected in the sound [...] but after observing it more, I think it is recorded"*). For some participants this assumption persisted even after the revelation moment (P8, P9: *"It's making some change but you just cannot figure out like what it reads to make this sound. No feedback"*), suggesting that the absence of clear, direct feedback overrode the explanation of the system's responsiveness.

In contrast, other participants preserved the idea of interactivity by hypothesising the presence of sensors despite the lack of evidence (P1, P6: *"There should be [sensors] [...] Why else would you have the big circuit thing? Because otherwise you could just put a speaker and play pre-recorded sounds"*) and by relying on visible cues (P7: *"[Parent] had the meter where you could see how loud it was. That's why I thought that might be an interactive element."*, P15: *"I saw some kind of [...] camera or a movement recorder or something and then I was like 'oh I think this kind of senses where I'm walking.'"*). It also emerged that for some participants, when the function of a component was not relevant or clear to their interpretations, it was understood as ornamental or directly dismissed in their sense-making of how the installation functioned (P2, P3, P18, P19, P7: *"[...] there was also just a lot of wires and plugs, and some of them weren't even doing anything, I felt. So I was like, is this just to add to the atmosphere, maybe?"*, P12: *"I think I definitely saw the laptop, but to me it was not obvious that that was part of the installation"*).

These responses suggest that when system behaviour is not clearly legible, some participants tended to construct or adjust coherent narratives to fill interpretive gaps in framing the installation's logic. A few inferred responsiveness from assumptions about technological complexity, suggesting that opaque computational systems may underlie sensing functions, while others expected legible functionality, interpreting visible components as cues

of responsiveness or, when their function remained unclear, dismissing them as ornamental or irrelevant in their sense-making.

4.2.3 Recognised More-than-Human Agency: Systemic Cooperation and Animistic Interpretations

A few participants (P5, P15, P17) explicitly attributed more-than-human agency to the installation after the revelation moment. Once the relationship between sonic behaviour, external stimuli and computational processing was exposed, this small group of participants was able to perceive the installation as a responsive system with its own distributed logic. The explanation shifted their understanding of the installation from an assemblage of sounding objects to an interconnected system exhibiting a form of purposive behaviour, where sound becomes the externalisation of invisible, responsive and unfolding cooperation rather than the output of separate and independent sources. For example, P17 explained: *"So like it was alive, like it is responding to something, the environment [...]"*. When the interview shifted towards the sonification of computational objects in an everyday imaginary, P10 recognised that they would perceive them *"as more alive"* too.

For P5, while the installation's distributed behaviour was not perceived as unsettling, imagining the same form of sonified responsiveness in a personal device led to discomfort due to an animistic interpretation (P5: *"I think I would find it a bit scary"*, *"I don't want it to be human. I don't want it to be like creating emotions based on what's going on"*). This contrast suggests that more-than-human agency may be more readily perceived when presented as a distributed network of interacting components in an artistic context, whereas a singular everyday device risks animistic interpretations, as P19 also reflected "*If I hear it in the exhibition, I think it's very interesting [...] but if I had to hear it 24-7, from something that I know has no capability of emotion, I don't think I would like it"*.

4.2.4 Unrecognised More-than-Human Agency: Legibility as Predictable Machinic Behaviour

Most participants did not explicitly recognise forms of more-than-human agency after the revelation moment, instead expressing increased curiosity towards the installation's technical functioning and interactivity. For example, P6 claimed*: "If you're able to influence, it makes me want to be like, oh, for certain, if I do this, this is going to happen. So it makes [...] me more curious if there are, like, certain actions I can take to influence the sounds"*. This can be interpreted through the notion of the Eerie affect [24]: when the "alien" logic behind an unfamiliar phenomenon is revealed, unsettlement diminishes as it becomes intelligible and thus predictable. In this sense, participants' perception of the installation shifted from that of an opaque and ambiguous phenomenon to an understandable system whose behaviour could be explained through linear mechanisms.

Although the revelation moment exposed hidden and cooperative processes, sparking a systemic perception for some (P4, P10, P11), such as P4, who claimed *"[...] they are part of a system, interacting with each other. My perception changes, I mean, there is something going on…"*, the installation was still largely interpreted as operating through a comprehensible logic, fostering operational curiosity rather than a broader perception of systemic more-than-human purposiveness.

4.2.5 Sound's Affective Role in Perceiving More-than-Human Agency

Although visual cues in the installation were relevant in instilling or reinforcing the ambiguity of interaction for some participants, the strong embodied attunement and absorption described by some suggests that sound, rather than constituting an interchangeable medium for representing more-than-human agency, also functions as an affective medium for sensing it, preceding conscious interpretation of a perceived phenomenon [35:81–84, 49]. For instance, the previously illustrated embodied experience of P3 described feeling that *"There's a certain type of beat that follows me"*, while P10 stated *"There was a strong feeling that this is, I mean, the present, the presence of something"* and P17 recalled feeling *"hunted and haunted at the same time"* and that "*there is something more than just us [...] something is going on"*. This suggests that while sound can signify hidden system activity in a

representational way, its encapsulating and temporal qualities also make such activity sensorially perceptible (e.g., P15: *"I think that it [sound] makes it easier to perceive the workings of it"*), allowing participants not only to recognise that the system is active, but also to sense its activity as an unfolding and immanent presence. In this sense, the sonification of computational inner processes does not merely signify agency, but also renders it perceptually accessible, enabling an entangled and experiential relationship with the interiority of its source, rather than an observed and distant acknowledgment of it [53, 67]**.**

This also emerges from two participants' reflections about how the role of sound was fundamental for their recognition of more-than-human agency in the installation (P15: "*It would make it seem more like a living being. Like how humans and animals make sounds based on what they're thinking or what they're feeling or how they're working*", P17: *"Sound, like for me, is connected more with life, with nature"*). In this way, sound allowed some participants to recognise more-than-human agency not solely through semantic associations with internal activity, but also to sense it as dynamic and purposeful.

4.3 Sonification and the Foregrounding of Computational Presence

Contingent Flows represented a design probe to invite participants to imagine the same type of sonification embedded in everyday devices. Most participants rejected this idea due to anticipated sensory overload, focusing mainly on sound aesthetics or seamless design solutions (P20, P3). For instance, P3 suggested a design approach in which device sounds blend with domestic environments, explaining that *"Maybe it would be nice to intentionally sonify in a way that's good for the environment. […] It just kind of merge into the background*".

Some participants recognised the potential usefulness of such sonification in everyday devices to raise awareness of complex internal processes and encourage more conscious use (P1), detect otherwise invisible information (P10), reveal computational complexity and capacity (P11, P13) and increase curiosity about how interaction affects system behaviour (P10, P17).

At the same time, it emerged that sonifying inner computational processes would heighten awareness of the presence of everyday technologies (P1, P19, P18: *"You would realize that technology is much more present than you think"*) and of its underlying data processing (P6, P7, P11: *"There is so much, and I'm not even aware of 10% of it […] if you get […] to know about it, I think this changes completely how you perceive it [referring to smartphones]"*). Particularly, P1 explained that *"The relationship between people and technology [...] We are currently doing it very subconsciously", "[There are] things that we're kind of giving more control to day by day...[Sonification] could be beneficial for actually being aware of what's going on",* suggesting that perceiving computational processes could foster greater awareness of unconscious reliance on technology due to the increased control delegated to it. P6 suggested that computational sonification could increase awareness of surveillance by making the scale of technological monitoring, sensing and data collection more perceptible: *"It might make us more aware of how much [...] of this [referring to technology] is surrounding us in our everyday life and how much data they're collecting, how much they're sensing our actions and the sheer size of it"*. This suggests that making computational activity audible could increase awareness not only of technological presence, but also of invisible processes of data extraction and monitoring.

However, while some participants valued this potential increased awareness in relation to privacy and control concerns (P6, P11) or to the habituated and unconscious use of personal devices (P1, P7), it was also often described as emotionally burdensome. Participants' concerns were frequently negotiated against a need for safety (P6) and for everyday serenity (P7, P9, P18), sometimes resulting in disinterest even when the relevance of these ethical matters was acknowledged (P12, P13). As P7 explained, *"maybe it's nice to be a bit, in a way, oblivious about it. I think what you said about the surveillance thing, it's maybe a bit scary if you think about it a lot"*.

Therefore, sonification foregrounds computational activity, amplifying in some participants pre-existing concerns about surveillance, data extraction and technological dependency. This suggests that habitual silence or

minimal sonification of personal devices contributes to comfort not only through sensory underload, but also by enabling engagement with technology without confronting the full extent of its operational presence. These findings indicate that sonifying computational processes may shape how participants perceive technology's presence, complexity and agency: a perception continuously negotiated in relation to the need for control, security and everyday comfort.

5 DISCUSSION

The study shows that sound can act as an affective mediator in shaping perceptions of computational objects by making their underlying activity perceptible and foregrounding their agentic presence, addressing the main RQ. SubRQ1 is addressed by demonstrating that posthumanist design approaches, here grounded in Fisher's theoretical lens [24, 25], can support the perception of more-than-human agency through the overturning of user-centred expectations of computational objects as passive and by rendering such agency experienceable rather than merely represented. SubRQ2 is addressed by showing that the research artefact can foreground latent expectations and tensions in interpretations of technological tools, while the Critical Design dialogical approach enables their further articulation and inquiry.

Although findings regarding the imagined sonification of everyday device processes are delimited by a hypothetical dimension, they revealed a valuable tension: increased awareness of computational presence emerged alongside concerns regarding privacy, surveillance and technological dependence. Therefore, the backgrounding of computational activity may contribute to keeping such worries latent by retaining technological objects' existence outside conscious perception. The study empirically aligns with pre-existing research [29, 33], showing how seamless design may support the concealment of the active co-participation of computational objects and how their minimised sonification can operate in an affective realm, favouring technology's integration in everyday life by backgrounding its existence, and thus making it less likely to be questioned.

However, this research also suggests that seamless interaction may remain preferable even when people recognise the ethical value of transparency. In fact, increased awareness of computational activity's extensiveness was often considered as burdensome due to the consequent foregrounding of individual pre-existing concerns, leading to an intentional preference towards the backgrounding of technological devices' underlying operations. The tension between the value of transparency and comfort emerges as a negotiation between the urge for awareness and the desire for safety and psychological ease, in which opacity might be accepted as the price for comfort. This suggests that transparency may not automatically constitute the preferred mode of relation with computational objects when it threatens convenience, extending Giaccardi and Redström's [33] argument by evidencing how seamlessness does not solely conceal computational agency, but may also function as a preferable affective condition to avoid confronting the pervasiveness, interconnection and opacity of everyday technological systems. This observation offers an ethical insight regarding transparency within HCI research. Besides understanding how to afford legibility of complex computational processes [1], it also becomes relevant to conceive forms of transparency that can enable preferred modes of co-existence with more-than-human technological entities without becoming psychologically overwhelming.

At the same time, the findings suggest that making devices' underlying activity perceptible through sonification may contribute to a strengthened sense of user agency by rendering the operational role of computational systems more intelligible. Some participants associated such sonification with increased awareness of technological presence, underlying data processing, surveillance and the unconscious, habituated use of devices. This suggests that the foregrounding of more-than-human agency may contribute to heightened intentional relationships with everyday technologies through a more evident and marked reconfiguration of human and more-than-human agency. By making the active co-participation of computational systems perceivable, users may be better positioned to engage with them more deliberately. In this sense, foregrounding more-than-human agential activity may paradoxically

support the reassertion of human agency by making the conditions of interaction more perceptible. This perspective gives practical relevance to the Entanglement Paradigm's [29] posthumanist approaches, suggesting that recognising distributed agency may enable more conscious, informed and fairly negotiated forms of human-technology relationships, thereby supporting greater ethical and deliberate relations with computational devices.

A further reflection concerns the role of sound in shaping perceptions of computational agency. The research shows how these perceptions are shaped by how computational activity is exteriorised and rendered perceptually accessible. The installation's sonic dynamism depended on design choices regarding which data to sonify and how to represent it, indicating that what is rendered as *Pheno-data* [47] is already mediated by human sense-making. At finer computational scales, sonification might have resulted in seemingly static noise due to the misalignment between human and computational temporal frameworks, still conceptually representing the installation's agential purposiveness while reducing its perceptible behavioural dynamism. This suggests that recognition of more-than-human agency also depends on being rendered at a scale compatible with human perceptual and cognitive resolution. Therefore, recognition of computational agency depends not only on foregrounded distributed processes, but also on the design mediation that makes them experientially sensible. Design configurations thus play an active role in determining whether computational agency remains backgrounded or becomes experientially discernible to people, highlighting HCI's ethical role in affording pre-mediated perceptions of technological devices. This reflection also stresses that this research's posthumanist approach, while challenging anthropocentric assumptions, still relied on human sensory and interpretive frameworks to make more-than-human agency perceivable.

The findings are limited by the study's situated nature, producing insights that emerge within specific sociomaterial configurations rather than being generalisable. While the results cannot claim that sonification produces recognition of more-than-human agency, they demonstrate that sound may create conditions for its emergence. Previous research suggests that engagement with invisible phenomena, here understood as the perceived agency of computational objects, is shaped by cultural and individual factors [36, 66]. Recognition of more-than-human agency may be further constrained by unfamiliarity with conceptual frameworks through which these experiences can be interpreted and articulated. The results are also shaped by an emotional and cognitive context of surprise and unease within the speculative study approach. The reliance on sound is an inherent design condition that privileges hearing-oriented forms of perception, suggesting future developments could explore additional modalities such as haptic feedback.

Additional limitations involve the unintentional invisibility of the laptop connected to the Parent artefact during the first 11 participants' encounters. Originally, 20 participants were recruited; P14's (informed participant) data were excluded from the analysis and are considered only here as contextual information. This constraint appears to have had limited impact, as only a minority of participants (P11, P14) attributed sense-making value to the laptop. Time constraints limited the implementation of additional sensors (see A.5). Although the system remained coherent with the concept design, this may have shaped the conditions under which agency and interactivity were perceived, potentially reducing the variability and responsiveness of the installation and thereby influencing participants' engagement.

The exhibition setting introduced uncontrolled variables during the observations, documented in the observation protocol's interference section. Opportunistic sampling may have influenced the diversity of perspectives captured. The research acknowledges that participants' affect and overall experience were shaped not only by the artefact but also by the physical and contextual setting of the exhibit [62].

The researcher's role as observer and discussion facilitator may have influenced participants' behaviour [7:33–38, 45] and responses, introducing potential interpretative bias due to single-researcher observation. This was addressed through discreet note-taking, brief interviewer notes (see A.3), structured observation protocol and audio-recorded interviews for re-listening. Paired visitors' behaviour and interviews may be shaped by interpersonal

dynamics and co-constructed sense-making [46], documented in the observation protocol and identified in the interviews' transcripts, which were considered during the data analysis.

Future research could explore how more-than-human agency is differently experienced depending on whether a system is depicted as distributed or as a single unified object. This direction would shed light on whether and how perceptions of agential purposiveness are shaped by how the distribution of agency is presented, inspired by P5's observed shift in agential perception (illustrated in section 4.2.3). Due to the speculative nature of the research, future work could phenomenologically explore how sustained exposure to sonified computational objects in everyday contexts might alter perceptions of more-than-human agency, awareness, habituation and acceptance, inspired by Wakkary et al.'s study [68]. Another inquiry could involve comparing how different sensory modalities (e.g., auditory, haptic and visual) may empirically differ in affording perceptions of more-than-human agency in computational objects.

AI USE

The thesis made use of the following AI-based tools: ChatGPT, Gemini and Claude.

ChatGPT was used for language support (grammar, syntax and lexical suggestions) and for exploring and comparing different qualitative analysis tools, contributing to the decision to employ Taguette and Figma.

Gemini was used to support the technical development of Python code in the first two conceptual prototypes and for coding experimentation with Arduino sensor circuits.

Claude was used for language support (grammar, syntax and lexical suggestions) and for identifying potentially relevant literature sources.

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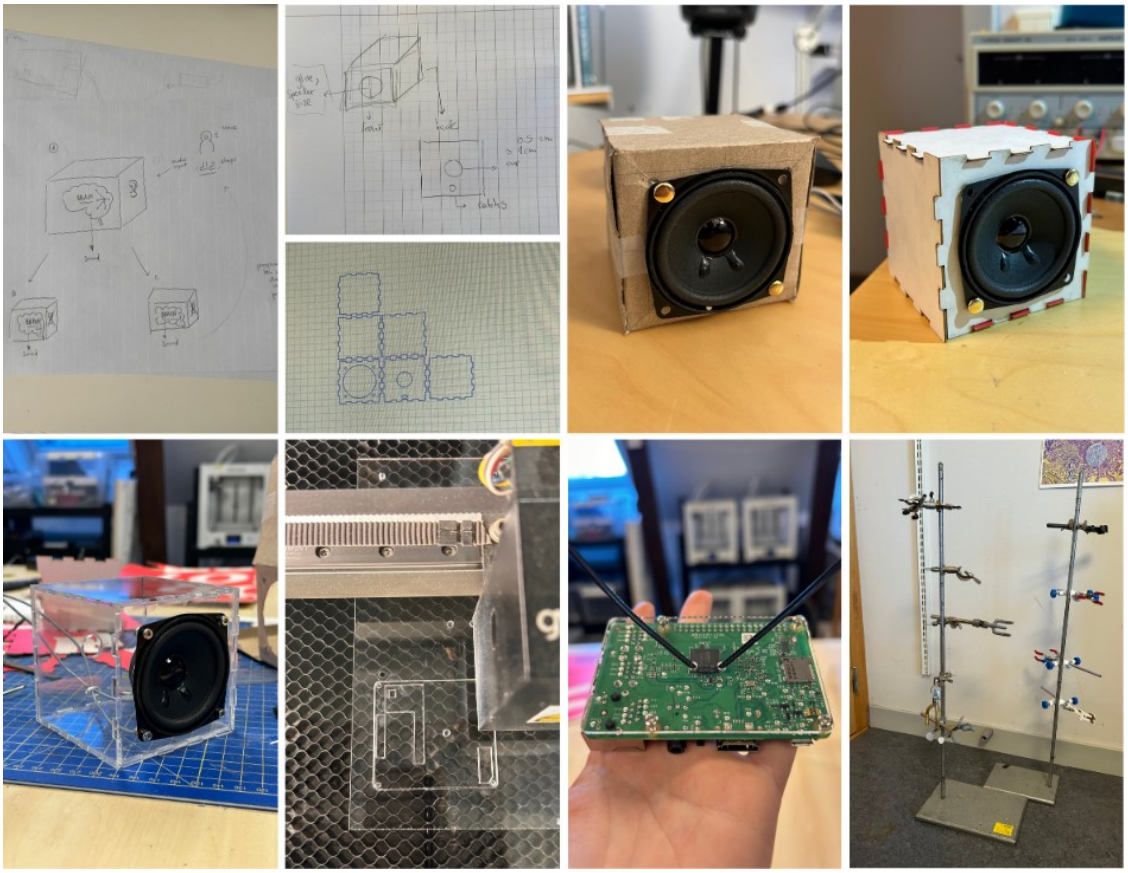
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A APPENDICES

A.1 Making process of the installation's artefacts



A.2 Observation Protocol

Date:
Participant ID:
Solo/paired: Solo □ Paired □ (if paired->separated IDs and note interactions between participants, same document)
Duration of encounter:

Section A - Observation field notes
This section is completed during the observation phase, when the participant(s) enter(s) the installation's room. Notes are taken chronologically and in real-time, considering the parameters outlined in Sections B, C and D.

- Overall quality of engagement (brief impressionistic assessment):
- Most significant moment of the encounter:
- Unexpected behaviours or reactions:
- Researcher's own observations and impressions:
- Notes on external factors that may have influenced the encounter (noise, other visitors, technical issues):

Paired participant notes *(complete only if two participants present together)*

Did one participant appear to lead the other? Yes ☐ No ☐

Notes on social dynamic between participants:

Were observed engagements with the installation notably different between the two (if yes, describe)? ☐ Yes ☐ No

Section B - Spatial and bodily behaviour

From this section to the following, the tables are compiled after the in-situ semi-structured interview with participants, leveraging the data gathered in Section A. Time is primarily intended as initial (1), during (2), final (3) engagements with the installation.

Behaviour	Time	Notes
Initial position and approach to the installation upon entering the space		
Proximity/distancing to parent/children object		
Body orientation (leaning in/away towards objects)		
Movement circling/exploring vs static observation		
Repeated return to specific object		
Touch/attempted touch of objects		
Attempts to interact with or influence the sound (body)		
Other:		

Section C - Facial and emotional cues

Cues	Time	Notes
Sustained attentive listening (stillness, eyes closed, absorption)		
Surprise (raised brows, pause in movement)		
Curiosity (leaning in, sustained gaze)		
Amusement or pleasure (smile, relaxed posture)		

Confusion (furrowed brows, head tilt)		
Discomfort or unease (stepping back, bothered/unsettled expressions)		
Indifference (minimal engagement, distracted gaze)		
Other:		

Section D - Verbal and non-verbal reactions

Reaction	Time	Notes (quotes/descriptions/non-verbal sounds)
Spontaneous verbalisation		
Exclamations or involuntary sounds (ex. laughing, puffing, hesitation markers)		
Silence (if notable/sustained)		
Attempts to interact with or influence the sound (voice)		
Comments directed at co-visitor (if paired)		
Questions directed to researcher		
Other:		

Possible further questions (just if the participant verbalises or has evident behaviours that can affirmatively answer to the questions):

- Did the participant notice sound responding to the environment?
- Did the participant attribute intention or agency to objects (verbal or gestural)?
- Did the participant anthropomorphise objects?

A.3 Interview Questions

part 1 (before)

- What is your background? Age?
- How would you describe what you just experienced?
- Was there a particular moment during the experience that struck you?
    - *if yes,* why?
- How would you describe the overall vibe or atmosphere?

part 2 - perception of the object (before)

- How would you normally describe or think about computational objects like this?
- What do you think is producing these sounds? (real-time/recorded)

Revelation moment – two options:

(a)If they identify the computational objects as sound source + its reaction to environmental changes

- Did that change/affect how you think about the object in front of you?

(b)If they don't identify the source-> revelation moment (*"the installation is sonifying its internal computational processes in real time, in reaction to environmental changes"*)

- What do you think of these objects now, knowing they are producing the sounds?
- Does knowing this change/affect anything for you and how you perceive them?
    - *if yes or no*, why? explore participant's impressions and thoughts

part 3 - broader reflection (after)

- We are surrounded by computational objects every day but most of the time they don't produce sounds that make perceivable what's going on inside them. How do you experience that in your everyday life?
- If your everyday devices sounded like this, how do you think that would affect your relationship with them?
- Would that be something you would welcome or not? Why?

Notes:

- Explore participants answers, try to engage with their perspective (Critical Design's dialogical methodology)
- Allow participants to develop their vocabulary to describe the encounter
- Ask all the questions while engaging in the conversation

Focus:

- Surprise, recategorisation, confusion, unsettling impressions;
- Spontaneous language used (ex. *anthropomorphisation*), comparison to familiar technology (*prior assumptions, subRQ2*);
- Attribution of intention/aliveness to the objects (*failure of absence*)
- Revise of an initial interpretation
- Dissonance between expectations and experience

A.4 Informed Consent Form

Study title: *Contingent Flows*: A Study of People's Experiences of a Sonic Installation

Researcher: Ginevra Bondesan (MSc student in Human-Computer Interaction, Uppsala University)

Contact: ginevra.bondesan.4728@student.uu.se

Purpose of the study: this study investigates how people experience and respond to a sonic installation. There are no right or wrong responses, the research is interested in your personal impressions, reflections and reactions to the encounter.

Thesis publication: participants can read the final thesis by searching for the researcher's name and surname on the DiVA Portal (www.diva-portal.org). The thesis is expected to be published in summer 2026, following final submission in June 2026.

What does participation involve? Your participation involves being observed during your first encounter with the installation and taking part in a following brief discussion (5-10 mins) about your experience and impressions. The interview will be audio recorded. The research doesn't expose participants to any psychological or physical risk or harm and it doesn't provide any compensation to participants.

Data processing: participants' personal data will be processed according to the EU General Data Protection Regulation (GDPR) law [2, 3:67–70].

What data will be collected? Written observational notes (1) taken during your encounter with the installation and audio recording (2) of the interview.

How will your data be used? Your data will be used for the purposes of this master's thesis research and may contribute to related future academic publications. Audio recordings will be transcribed using the European Commission's Speech-to-Text tool [1]. All gathered data will be pseudonymised using a separate participant ID code document which only the researcher has access to, this particular document will be deleted by June 2026 (expected graduation date). You will not be identifiable in any public output (thesis, presentations, future research articles, etc.) of the research. Quotes from your interview may be used in publications, with all identifying details removed.

How will your data be stored? Recordings and notes will be stored securely on a password-protected personal device. Data will be retained for the duration of the research project and deleted upon completion, unless the data contributes to subsequent academic publications, in which case it will be retained for 10 years following the last publication derived from the research.

Your rights:

- Participation is entirely voluntary
- You have the right to request access to your personal data at any time prior to the deletion of the ID code document (deadline: June 2026, expected graduation date) by contacting the researcher (email contact above)
- You may withdraw from the study at any point during the interview without giving a reason
- You may request that your data be withdrawn up until 24/03/2026, before the research analysis process starts, by contacting the researcher (email contact above)

Consent:

By signing below I confirm that:

- I have read and understood the above information
- I agree to participate in the study and to the processing of my data as described above
- I agree to the audio recording of my interview
- I understand I can withdraw my data until 24/03/2026

Name: ……

Signature: …… Date: .....

I confirm I am 18 years of age or older: ☐ Yes

Sources:

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A.5 Tested sensors not included in the final installation

The pictures show the initial conceptual prototyping phase, involving testing different Arduino circuit configurations with temperature-humidity, light and sound sensors. The final installation would also have included the temperature-humidity and the ambient light sensors, which were not integrated due to time constraints. The final installation therefore featured camera and microphone sensors but not the Arduino sensor system presented here.

